Machine Learning in Empirical Asset Pricing

Compact Vocabulary Cheat Sheet • companion to Gu, Kelly & Xiu (2020)

1. Core Finance Concepts

Risk Premium. Extra expected return for holding a risky asset vs. the risk-free rate. Mechanically earned via a lower purchase price today.

Expected Return. Probability-weighted average of future returns; the central quantity asset pricing aims to measure.

Cross-Section. Many assets compared *at the same time*. "Cross-sectional prediction" = ranking stocks against each other, not forecasting the market overall.

Time Series. One asset followed *across time*. Complementary slice of the data.

Systematic vs. Idiosyncratic Risk. Systematic risk is market-wide and earns a premium; idiosyncratic risk is asset-specific and diversifies away (no premium).

Sharpe Ratio. Return per unit of risk: $(\bar{r} - r_f)/\sigma$. Market ≈ 0.4; strong quant strategy ≈ 1.0.

Standard Deviation of Returns. Typical size of the deviation from mean return. Not the mean itself — it measures wobble, not level. See §6 for the exact formula.

Momentum. Empirical tendency for stocks with strong recent returns (6–12 months) to keep outperforming short term.

Liquidity. Ease of trading without moving the price. Illiquid stocks earn a premium as compensation.

Volatility. Magnitude of price fluctuations. Used as both a risk measure and a predictor.

2. Machine Learning & Data

Machine Learning. Methods that learn patterns from data instead of assuming a fixed functional form. Valuable for capturing nonlinearities and interactions.

Big Data. Datasets large in both observations and features. ML stays stable at scales where classical regression breaks down.

Return Prediction. Forecasting future returns from current information. The operational task of empirical asset pricing.

Feature / Predictor. An input variable (size, past return, book-to-market, etc.).

Overfitting. Model fits noise, not signal. Out-of-sample performance collapses. Regularization is the main defense.

Cross-Validation. Splitting data into folds to tune hyperparameters without peeking at the test set.

Regularization. Adding a penalty to the loss to discourage complex models. Ridge, Lasso, Elastic Net are canonical.

Hyperparameter. A knob controlling model complexity (e.g. λ, tree depth). Tuned via cross-validation, not fitted.

3. Penalized Linear Methods

Ridge Regression. L_2 penalty; shrinks coefficients but rarely zeros them. Stabilizes fits under correlated predictors.

Lasso. L_1 penalty; drives some coefficients exactly to zero $\Rightarrow$ automatic variable selection.

Group Lasso. Predictors partitioned into groups; each group is either entirely in or entirely out. Penalty is L_1 *across* groups, L_2 *within*.

Elastic Net. Convex combination of Ridge and Lasso penalties. Selection from Lasso, stability from Ridge.

Coordinate Descent. The algorithm behind `glmnet` and `sklearn`: update one coefficient at a time via soft-thresholding.

Dimension Reduction. Compress many correlated predictors into a few composites before regressing. PCR and PLS are standard.

4. Tree-Based Methods

Decision Tree. Recursive yes/no splits on predictor values. Captures nonlinearity and interactions; overfits alone.

Random Forest. Average many trees, each trained on random subsets of rows and columns. Variance reduction through averaging.

Gradient Boosting. Train trees sequentially, each correcting residuals of the last. XGBoost, LightGBM. Top performer in the paper.

Ensemble. Any combination of simpler models. Forests are parallel; boosting is sequential.

5. Neural Networks

Neural Network. Stacked layers of weighted sums + nonlinear activations. Approximates complex nonlinear functions given enough data.

Deep Learning. Neural networks with many hidden layers. More depth $\Rightarrow$ richer representations, harder training.

Activation Function. The nonlinear squish inside each neuron (ReLU, tanh, sigmoid). The source of nonlinearity.

Backpropagation. Gradient-based training algorithm; computes partials of the loss w.r.t. each weight efficiently.

Dropout / Batch Norm. Standard regularization and stabilization tricks for deep nets.

6. Evaluation & Portfolio Metrics

Out-of-Sample R^2. Variance in future returns explained by past-data predictions. Even 0.5–1% monthly is meaningful.

Long–Short Portfolio. Buy top-predicted stocks, short bottom-predicted. Isolates forecasting skill from market direction.

Decile Sort. Rank stocks into 10 bins on a signal; compare mean returns. A simple diagnostic.

Annualized Sharpe. Monthly Sharpe $\times\sqrt{12}$ under rough independence. Standard reporting unit.

Variable Importance. Per-feature contribution to forecasts. Gu–Kelly–Xiu: momentum, liquidity, volatility dominate in every method.

7. Context & Applications

Fintech. Financial technology — robo-advisors, algo-trading, AI credit scoring. ML return prediction is a core input.

Quant Finance. Investing via statistical models rather than discretion. Main industrial application of this research.

Anomaly / Factor. Firm characteristic linked to predictable return differences. The "factor zoo" has hundreds; ML separates signal from noise.

Mispricing vs. Risk. Ongoing debate: are return patterns risk compensation or investor mistakes? Evidence supports both.

Key Formulas at a Glance

Return & Risk Measures

Mean monthly return:

$$\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$$

Monthly std. deviation:

$$\sigma_{\text{mo}} = \sqrt{\tfrac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2}$$

Annualization (i.i.d. assumption):

$$\bar{r}_{\text{ann}} \approx 12\,\bar{r}, \qquad \sigma_{\text{ann}} = \sqrt{12}\,\sigma_{\text{mo}}$$

Sharpe ratio:

$$\text{Sharpe} = \frac{\bar{r} - r_f}{\sigma}$$

Portfolio Variance

For weights $w = (w_1, \ldots, w_N)'$ and covariance matrix Σ:

$$\sigma_p^2 = w'\Sigma\,w = \sum_i w_i^2\sigma_i^2 + 2\sum_{i<j} w_i w_j \rho_{ij}\sigma_i\sigma_j$$

Diversification: when $\rho_{ij} < 1$, off-diagonal terms are small, and $\sigma_p < \sum_i w_i\sigma_i$.

Two-asset, equal weights, each $\sigma = 20\%$:

- $\rho = 1$: $\sigma_p = 20\%$ (no benefit)
- $\rho = 0$: $\sigma_p \approx 14.1\%$
- $\rho = -1$: $\sigma_p = 0$ (full hedge)

Penalized Linear Objectives

Ridge: $\sum_i (y_i - x_i'\beta)^2 + \lambda \sum_j \beta_j^2$

Lasso: $\sum_i (y_i - x_i'\beta)^2 + \lambda \sum_j |\beta_j|$

Group Lasso:

$$\sum_i (y_i - x_i'\beta)^2 + \lambda \sum_g \sqrt{p_g}\,\|\beta^{(g)}\|_2$$

Elastic Net:

$$\sum_i (y_i - x_i'\beta)^2 + \lambda\left[\alpha\sum_j |\beta_j| + (1-\alpha)\sum_j \beta_j^2\right]$$

Reading the Formulas

y_i: return to predict. x_i: predictor vector.
β: coefficients. λ: penalty strength.
$\alpha \in [0,1]$: elastic-net mix (1=Lasso, 0=Ridge).
$\beta^{(g)}$: coefficient sub-vector for group g.
w: portfolio weights. Σ: return covariance.
ρ_{ij}: correlation of returns i and j.